



Review

Systemic Nickel Allergy Syndrome[☆]Allison Freeman, BA^a, Ashleigh Letterman, BS^a, Jeffrey Boos, MD^b, Scott Jackson, MD^{a,*}^a Alice L. Walton School of Medicine, Bentonville, Ark^b Brigham and Women's Hospital, Boston, Mass.

ARTICLE INFO

Keywords:

Allergy
Dermatitis
Diet
Fibromyalgia
Irritable bowel syndrome
Nickel

ABSTRACT

Systemic nickel allergy syndrome (SNAS) is characterized by both cutaneous and extracutaneous symptoms following systemic exposure to nickel, most commonly through the ingestion of nickel-containing foods in individuals sensitized to nickel via skin contact. Patients with SNAS present with dermatitis in addition to gastrointestinal, neurologic, and other extracutaneous symptoms. The most widely implemented treatment is the low-nickel diet. The present authors saw a patient with a history of allergic contact dermatitis to nickel experience resolution of hand dermatitis, chronic pain (diagnosed as fibromyalgia), and chronic diarrhea (diagnosed as irritable bowel syndrome) with a low-nickel diet after just 1 month. She was diagnosed with SNAS, and this intriguing case prompted the review that follows.

Systemic nickel allergy syndrome (SNAS) is a consequence of systemic exposure to nickel in individuals previously sensitized to nickel via skin contact. The term first appeared in the medical literature in 2011.¹ In contrast to allergic contact dermatitis (ACD), which presents with localized dermatitis in response to direct cutaneous contact with nickel, SNAS is a systemic reaction after ingestion of nickel-rich food or in the presence of some other internal source of nickel.² Patients with SNAS present with cutaneous signs and symptoms typical of the so-called systemic contact dermatitis in addition to gastrointestinal, neurologic, and other extracutaneous symptoms.³ Interventions for the management of SNAS include low-nickel diet, nickel desensitization, nickel chelation, systemic medications, and probiotics.

Pathogenesis of SNAS

Much of what we know about the pathophysiology of nickel hypersensitivity comes from our understanding of allergic contact dermatitis. In ACD from nickel, nickel ions must penetrate into the viable portion of the epidermis to affect the patient's immune system. The penetrating dose of nickel is determined by the concentration ($\mu\text{g}/\text{cm}^2$) of nickel, which varies by the material in contact with the skin, as well as by the condition of the skin in the place of contact, as determined by the presence of friction or occlusion, moisture or lack thereof, acidity, humidity, age of patient, anatomic site, and so forth. Barrier dysfunction in the setting of atopic dermatitis or irritant contact dermatitis facilitates nickel penetration. Duration of contact largely correlates with the likelihood of reaction, but recent studies by Erfani et al and Ahlstrom et al reveal

that skin deposition of nickel can occur after seconds to minutes of contact.^{4,5} Once the nickel has reached the viable epidermis, delayed-type hypersensitivity can ensue. In sensitization, nickel ions bind to toll-like receptor 4 on dendritic cells and activate them. These dendritic cells migrate to the local lymph node and present the nickel to naive T cells. Memory T cells specific for nickel proliferate and migrate to the skin, and the patient is now sensitized. A future exposure could activate these cells, leading to what is known as elicitation. The skin-residing memory T cells, activated by new exposure to nickel ion, recruit other memory T cells, and all of them together cause inflammation and epidermal damage, which clinically manifests as allergic contact dermatitis. If the patient is exposed to nickel internally, these same memory T cells become activated, triggering systemic symptoms such as gastrointestinal symptoms as well as cutaneous reactions, including dermatitis at previous sites of exposure. Thus, the pathogenesis of SNAS predominantly involves a type IV (delayed-type) T cell-mediated hypersensitivity pathway. It is important to know that some patients with SNAS can have a type I hypersensitivity presentation, in which the nickel exposure causes urticaria or angioedema.

Nickel Allergy

Nickel hypersensitivity is most commonly diagnosed by patch testing. Nickel sulfate hexahydrate in petrolatum is applied by patch to the skin and left in place undisturbed for 48 hours. The patch is then removed and the skin is inspected at 48 hours and 96 hours for erythema,

[☆] Funding: None.

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edema, and blisters on a 4-point scale. Concentrations of 2.5% and 5% are used most commonly in North America and Europe, respectively. Nickel is the second most common cause of contact allergy behind the toxicodendron plants (poison ivy, oak, and sumac), and it is the most common allergen to induce a positive reaction on patch testing. According to Warshaw et al, the average frequency of positive patch test results to nickel was 17.5%, with the prevalence of nickel sensitivity increasing from 14.3% in 1994-1996 to 20.1% in 2013-2014.⁶ In other studies, 16%-18.2% of US patch test patients reacted to nickel,^{7,8} and 22.4%-24.3% of Canadian patients tested positive to nickel.⁹ Nickel allergy remains an issue in Europe also, in spite of the Nickel Directive implemented in 1994, which did lead to a decrease in incidence of nickel allergy. The prevalence of nickel allergy in the European general population is approximately 8%-19% in adults and 8%-10% in children and adolescents, with a strong female predominance.¹⁰ The results of patch testing for nickel allergy vary by the concentration of nickel used in the patch. Svedman et al showed that contact allergy to nickel 5.0% and 2.5% concentrations was elicited in 20.3% and 16.8%, respectively ($P < .0001$).¹¹

Adults have generally been found to exhibit nickel allergy more commonly than children. Women have been found to have a higher prevalence of nickel allergy than men, possibly because of higher levels of metal exposure in jewelry and cosmetic products. According to Conti et al, 22% of women compared to 5.3% of men have nickel allergy.¹² Hairdressers, caterers, cleaners, and metalworkers also have higher rates of nickel allergy because of their higher exposure to nickel compared to other professions.¹² Specific populations may also be more susceptible to nickel allergy than others because of dietary variations affecting nickel intake. Other risk factors for the development of nickel allergy include atopic dermatitis and the presence of dental metals and various surgical implants.

Direct skin contact with nickel-containing materials induces delayed-type hypersensitivity, which presents as contact dermatitis. In most cases, nickel dermatitis is subacute or chronic dermatitis. Itch is a major feature of allergic contact dermatitis. Lesions consist of scaly, fissured, crusty papules coalescing into plaques. Chronic lesions are often lichenified. Nickel dermatitis presents most commonly on the earlobes in patients with earring allergy, around the umbilicus in areas in contact with nickel-containing clasps or buckles, and around the wrist in nickel-containing watches. Questioning the patient about a history of nickel allergy based on this clinical information can be a helpful starting point in the evaluation of nickel allergy, but patient-reported history has less validity than one might expect. Josefson et al reported that asking the questions "Are you sensitive/hypersensitive/allergic to nickel?" and "Do you get a rash from metal buttons, jewelry or other metal items that come in direct contact with your skin?" have a positive predictive value of only 59% and 60%, respectively¹³; 40% of the study patients answered "yes" to the first question and 35% answered "yes" to the second question, but only 30% tested positive for nickel on patch testing. Thus, questioning the patient will lead to overestimating of nickel allergy, and therefore patch testing is the best method for confirming this allergy.

Systemic Exposure to Nickel

Nickel is abundant in the environment and is found in a variety of foods, making avoidance challenging. Most dietary nickel comes from plant-based foods, raising concerns because avoidance of nickel-rich foods may lead to a diet low in fruits and vegetables. Common nickel-rich foods include chocolate, leafy green vegetables, legumes, beans, soy, nuts, whole wheat, peas, oats, shellfish, and canned food.¹⁰ Drinking water is another source of dietary nickel; in European countries, the European Union regulates the maximum amount of nickel in drinking water to 20 µg/L. Notably, Ahlström et al reported a case in which water containing 53 µg/L of nickel triggered facial dermatitis in a patient. Importantly, the amount of dietary nickel consumed may vary based

on factors such as soil composition, environmental pollution, and water quality.¹⁴ In addition to dietary sources, numerous environmental factors may contribute to systemic nickel exposure. Polluted urban environments pose an additional risk because traces of nickel are found in traffic-related air pollution. Everyday consumer products such as jewelry, metal tools, cosmetics, metallic pieces on clothing, coins, electronic devices, and cooking utensils are also possible sources of exposure that may trigger cutaneous and systemic symptoms. Even drinking liquids out of stainless-steel water bottles can reportedly lead to nickel exposure.¹⁴ Dental and medical devices such as permanent retainers and endovascular implants pose a risk because they may contain traces of nickel. Patients with nickel allergy may experience adverse outcomes following the implantation of a nickel-containing endovascular device.¹⁵ Kocayigit et al reported a case of a systemic allergic reaction to nickel in an atrial septal defect closure device that necessitated surgical removal of the device.¹⁶ Nosbaum et al reported on a nickel-induced systemic dermatitis from a sacral neurostimulator.¹⁷ Kounis syndrome is acute coronary syndrome in the setting of a hypersensitivity reaction, and metal allergy (including nickel allergy) can lead to Kounis syndrome in patients with coronary stents and atrial septal occluder devices.^{18,19} Both of the FDA-approved devices for patent foramen ovale (PFO) closure are constructed from nitinol, a nickel-titanium alloy, raising concern for adverse effects in patients with nickel hypersensitivity. In a double-blind study evaluating whether nickel hypersensitivity is associated with adverse outcomes following PFO closure, Apostolos et al found that patients with nickel allergy reported a significantly higher proportion of adverse effects compared with those without nickel hypersensitivity.²⁰ The most commonly reported symptoms were new-onset or worsening migraines and palpitations. Overall, nickel allergy was associated with a 10.5-fold increased risk of adverse effects following PFO closure. Notably, the reported adverse events were relatively mild, and patients were generally able to tolerate the implanted devices without clinically significant reactions. Within 90 days of PFO closure, there were no reported cases of death, bleeding, transient ischemic attack, or stroke. These findings have prompted discussion regarding the clinical significance of nickel hypersensitivity in PFO closure and whether reported symptoms reflect true hypersensitivity reactions or represent transient, multifactorial postprocedural effects. Cole et al reported a case of a systemic allergic response to a nickel-containing foreign body (a BB gun pellet) lodged in the orbit of a 12-year-old boy with a history of nickel allergy.²¹ McLean et al reported on a generalized allergic reaction in a toddler who ingested nickel-containing metal coins.²²

Symptoms of SNAS

The symptoms of systemic nickel allergy syndrome can be categorized into the following 4 categories: gastrointestinal, dermatologic, rheumatologic, and neurologic. Some patients will have only 1 or a few of these symptoms while others might have many. Gastrointestinal symptoms were the most common in a study published by Ricciardi et al.²³ According to Ricciardi et al, in a study of 98 patients with systemic nickel allergy syndrome, upon oral challenge with 1.25 to 3.75 mg of nickel sulfate, meteorism and bloating were the 2 most common symptoms, affecting 67.3% and 57% of patients, respectively. Heartburn and recurrent abdominal pain occurred in 37.6% and 30.6% of patients, respectively, and diarrhea and constipation each in 16.3% of patients. De Waal recently reported a case of a patient with dermatologic nickel allergy who experienced gastrointestinal symptoms and received a diagnosis of irritable bowel syndrome.²⁴ Her intestinal and extraintestinal symptoms either resolved or were better managed after she started a low-nickel diet. Lactose intolerance is more common in patients with SNAS. In a study by Cazzato et al, among 178 patients with SNAS and 60 healthy controls, positive lactose breath test occurred in 74.7% of the SNAS group compared with 6.6% of the control group.²⁵ The authors hypothesized that nickel-induced inflammation could temporarily impair the brush border enzymatic functions, resulting in hypolactasia. Finally,

Stanghellini et al conducted a study that demonstrated a higher prevalence of nickel sensitivity among individuals with gastroesophageal reflux disease (GERD) compared with non-GERD patients, suggesting a role for nickel-induced immune activation that may contribute to gastrointestinal symptoms.²⁶ This clinical overlap highlights the potential for nickel hypersensitivity to present with extra-cutaneous manifestations that may be misattributed to a strictly gastrointestinal disorder, further supporting the concept of SNAS as a systemic immune-mediated condition rather than an isolated contact allergy.

Dermatologic manifestations of systemic exposure to nickel in the patient allergic to nickel are numerous. The best known is the vesicular hand and/or foot dermatitis that mimics dyshidrotic eczema. Less common is widespread dermatitis or generalized pruritus without dermatitis. Zirwas et al noted symmetrical dermatitis over the extensor elbows as a clue.²⁷ Thirty-three percent of patients in the Ricciardi study experienced exacerbation of allergic contact dermatitis, and systemic contact dermatitis occurred in 15.3%.²³ Urticaria and angioedema were also documented, with urticaria and angioedema manifesting in 23.5% of cases. Silvestri et al reported a case of pruritus ani caused by nickel allergy in the setting of daily peanut butter consumption that resolved with dietary nickel restriction.²⁸ This cutaneous presentation of nickel allergy might represent a variation of the baboon syndrome, which is a flexural or intertriginous inflammation most often discussed as a type of reaction to medication. Kolodziej et al reported a case of baboon syndrome after the systemic administration of nickel.²⁹ Mancuso et al reported a case of eyelid dermatitis and conjunctivitis as only manifestations of allergy to nickel in an orthodontic appliance.³⁰ Other forms of systemic contact dermatitis include dermatitis at sites of previous exposure to the allergen or at sites of previous positive patch tests, urticarial dermatitis, exanthem, erythroderma, and vasculitis-like lesions.

The neurologic symptoms of systemic nickel allergy syndrome are less common. In the Ricciardi study, headache was the most common neurologic symptom (33.7% of cases), with migraine headache occurring in 7.1%.²³ Dizziness and chronic fatigue are other symptoms associated with this syndrome. However, in a study by Braga et al, headache, dizziness, and fatigue, although reported present after nickel-rich meals by the study patients, could not be elicited by oral nickel challenge, the gold standard for diagnosis systemic nickel allergy.³¹ Fernandes et al reported a case of severe migraine associated with nickel allergy that required surgical removal of an atrial septal device.³²

While chronic pain and fatigue are nonspecific but commonly reported symptoms, rheumatologic and autoimmune manifestations appear to be relatively uncommon in systemic nickel allergy syndrome. Loyo et al reported a case of a woman with high fever, extreme fatigue, transient thrombocytopenia, multiple cervical adenopathies, hepatosplenomegaly, anemia, neutropenia, severe proteinuria and urine sediment abnormalities, elevated serum ferritin levels, and transient low positive antinuclear antibodies 1 year after she had a nickel–titanium chin implant inserted for cosmetic reasons.³³ All clinical findings resolved when the implant was removed. The authors contend this case represents a newly described diagnosis called ASIA syndrome, which stands for autoimmune/autoinflammatory syndrome induced by adjuvants. Stejskal et al showed that all 15 of their patients with fibromyalgia vs 10 control patients without fibromyalgia had a least 1 metal allergy vs none in the control group.³⁴ Nickel allergy was the most common. Removal of metals or reduction in metal exposure led to resolution of symptoms in 70% at 5 years.

Pulmonary symptomatology is the least discussed category in the medical literature, but SNAS may present with asthma and rhinitis.³⁵ Kolberg et al reported that wheezing and asthma were associated with self-reported nickel allergy in young German adults.³⁶ But in the study authored by Ricciardi et al, no patient reported respiratory symptoms.²³

From an endocrinologic standpoint, nickel contact allergy has been correlated with obesity, and there is speculation over a role in HPA axis impairment. Watanabe et al's study of 1128 overweight and obese outpatients showed that nickel allergy is more common in this popula-

tion and is associated with worse metabolic parameters and impaired growth hormone secretion.³⁷ Lusi et al reported that the prevalence of nickel allergy in a population of 72 overweight female patients was 59.7%, compared with 12.5% in the general population.³⁸ Interestingly, a significant reduction in body mass index was observed in 55.8% of the overweight females with nickel allergy after 24 weeks on a low-nickel diet. Risi et al reported an association between SNAS and growth hormone–insulin-like growth factor axis dysregulation in obese individuals.³⁹ They patch tested 859 subjects with obesity and suspected nickel allergy. The subjects with nickel allergies showed a greater GH-IGF1 axis impairment, a higher prevalence of empty sella (ES), a reduced pituitary volume, and a higher normalized T2 pituitary intensity compared with nonallergic ones. Andrioli et al found that SNAS patients have a higher prevalence of expressing thyroid autoimmune markers, which may increase their risk of developing an autoimmune thyroid disorder such as Hashimoto thyroiditis. This study suggests that SNAS is more than just a localized contact allergy but a systemic immune activation that may progress to further autoimmune conditions.⁴⁰ Yuk et al conducted a study of 4985 women and identified a significant association between nickel allergy and endometriosis, with patients with nickel allergy exhibiting a 2.47-fold increased odds of having endometriosis.⁴¹ The authors propose that the estrogenic properties of nickel, in combination with increased nickel exposure and elevated systemic nickel levels in women, may contribute to the development of estrogen-dependent conditions such as endometriosis. However, they emphasize that these findings should be interpreted cautiously and that further investigation is needed to clarify their clinical significance.

With these varied symptoms (Table 1) across multiple organ systems in mind, the differential diagnosis of systemic nickel allergy syndrome is broad and encompasses several seemingly unrelated diagnoses. It includes irritable bowel syndrome, chronic dermatitis, chronic headaches, fibromyalgia, and even pituitary axis dysregulation. We have seen resolution of hand dermatitis, chronic pain, and chronic diarrhea in a patient placed on a low-nickel diet after just 1 month. The prevalence of SNAS among the patients with these diseases is not known. But with a prevalence of nickel allergy as high as 10%-20%, further study of populations of patients with conditions such as fibromyalgia and IBS, as well as assessment of the role of nickel allergy in these populations, is warranted. However, routine testing for nickel allergy in IBS patients is not presently recommended by the American College of Gastroenterology.

Diagnosis of SNAS begins with a thorough clinical history documenting ACD.¹² A positive nickel patch test is nearly always part of the diagnosis but would not necessarily indicate systemic involvement. Reduction in symptoms while following a low-nickel diet is considered diagnostic of SNAS. After 4-6 weeks on a nickel-free diet, patients may undergo the oral provocation test, commonly known as the oral nickel challenge, which is considered the gold standard for SNAS diagnosis.¹² Diagnosis is confirmed if consumption of progressively increasing concentrations of nickel under controlled conditions reproduces the patient's systemic symptoms.⁴²

Treatment of Systemic Nickel Allergy Syndrome

The treatment of systemic nickel allergy syndrome can be divided into 5 categories: a low-nickel diet, removal of the nickel-containing implant or material from the body, nickel hyposensitization measures, treatments that mitigate the allergic response to the presence of nickel, nickel chelation, and probiotics. A low-nickel diet refers to the exclusion of nickel-rich foods and is currently recognized as the standard treatment for patients with SNAS. Symptom improvement also varies based on patient adherence and their baseline nickel sensitivity. Foods rich in nickel (Table 2) are reduced in the diet or altogether avoided. Data from a recent study suggest that nickel contamination in poultry, beef, and pork has a minimal impact on human exposure, posing a negligible risk to public health.⁴³ De Mata et al reported that 60 days on a low-nickel diet led to a decrease in cutaneous and extracutaneous symptoms.⁴⁴

Table 1
Potential Clinical Manifestations of Systemic Nickel Allergy Syndrome (SNAS).

Gastrointestinal	Meteorism; bloating; heartburn; recurrent abdominal pain; diarrhea; constipation; lactose intolerance; irritable bowel syndrome, gastroesophageal reflux disease (GERD)
Dermatologic	Vesicular hand and/or foot dermatitis (dyshidrotic eczema-like); exacerbation of allergic contact dermatitis; systemic contact dermatitis; baboon syndrome; eyelid dermatitis; pruritus ani; widespread dermatitis; generalized pruritus; symmetric extensor elbow dermatitis; urticaria; angioedema; dermatitis at sites of previous exposure to the allergen or at sites of previous positive patch tests, urticarial dermatitis, exanthem, erythroderma, and vasculitis-like lesions.
Neurologic	Headache; migraine headache; dizziness; fatigue
Rheumatologic	Chronic pain; fibromyalgia-like symptoms; fatigue; ASIA syndrome
Respiratory	Asthma; rhinitis; wheezing
Endocrinologic	Obesity; HPA axis disruption; reduced pituitary volume; autoimmune thyroid disease; endometriosis

Table 2
Nickel-Rich Foods to Restrict on the Low-Nickel Diet.

- **Cocoa and chocolate** (especially dark chocolate, cocoa powder)
- **Soybeans and soy products** (tofu, soy milk, soy flour)
- **Legumes** (lentils, chickpeas, peas, kidney beans)
- **Nuts** (cashews, peanuts, hazelnuts, almonds)
- **Seeds** (sunflower seeds, sesame seeds)
- **Whole grains** (whole wheat, oats, bran, buckwheat)
- **Oatmeal**
- **Shellfish** (especially mussels and oysters)
- **Canned foods** (nickel can leach from the can into food)
- **Certain vegetables** (spinach, kale, asparagus, broccoli)

After 60 days, patients who reintroduced nickel experienced a recurrence of symptoms while those who maintained the low-nickel diet continued to show a reduction in symptoms. Rizzi et al demonstrated a marked improvement in 20 IBS patients with suspected SNAS placed on a low-nickel diet. Despite the effectiveness of a low-nickel diet, it has notable limitations.⁴⁵ Nickel-rich foods are abundant, making adherence difficult to maintain long term. The broad dietary restrictions may negatively affect quality of life and reduce compliance if not carefully managed with individualized planning and nutritional support.¹² The severity of restrictions in a low-nickel diet may lead to a number of adverse consequences, including nutritional deficiencies, reduced fiber intake, and inadequate intake of essential vitamins and minerals, especially in patients with gastrointestinal comorbidities who already face dietary limitations.⁴⁶ Nutritional counseling has therefore been explored as a supportive strategy to mitigate the challenges associated with a low-nickel diet. In 1 study, SNAS patients who received telehealth nutritional counseling that included devising an individualized diet plan, assessment of nutritional status, and video consultations demonstrated improved quality of life and greater adherence to low-nickel diet plans after 30 days.⁴⁶ Another strategy to mitigate this is the use of hydroponic cultivation to reimplement nickel-rich foods. One clinical trial demonstrated that the use of hydroponic-cultivated tomato sauce in combination with a low-nickel diet in those with SNAS improved their quality of life and SNAS-related gastrointestinal symptoms.⁴⁷

The use of oral immunotherapy (OIT) for nickel hyposensitization has also been investigated. Oral nickel hyposensitization refers to the gradual, controlled oral administration of nickel in microdoses to reduce hypersensitivity and improve tolerance. Clinical studies suggest that nickel hyposensitization is an effective immunotherapy that may allow future reintroduction of nickel into diet without symptoms. A controlled clinical trial by Di Gioacchino et al in 2014 included 112 SNAS patients who were randomized to receive either microdoses of oral nickel sulfate or a placebo for 1 year while maintaining a low-nickel diet.⁴⁸ The patients receiving nickel sulfate had gradually increasing doses up to 1.5 µg/week. The results showed that patients who received oral nickel immunotherapy had a significant improvement in cutaneous and gastrointestinal symptoms associated with nickel allergy compared with those who received the placebo. The study reported that patients also had improved tolerance of nickel reintroduction into diets following OIT for 1 year, allowing a less restricted diet. A clinical trial by Minelli

et al in 2010 studied the effects of OIT for nickel hyposensitization in patients with nickel allergy.⁴² Patients were randomized to receive either nickel sulfate or placebo, both groups adhering to a low-nickel diet. Twenty-three patients received OIT over the course of 10 weeks in increasing doses (from 0.1 ng to 3 µg) and then received a maintenance dose of 1.5 µg/week for 1 year. Twelve patients received a placebo. The reintroduction of dietary nickel occurred after 4 months into OIT. The results showed that following the study, levels of IFN-γ, IL-13, and IL-5 in all patients receiving treatment significantly decreased compared to patients receiving placebo, suggesting improved immune tolerance. Twenty out of 23 patients receiving OIT successfully tolerated gradual reintroduction of dietary nickel compared to none in the control group. Limitations remain for oral desensitization, including the lack of standardized dosing protocols for OIT of nickel. Longer and larger trials are needed, as information over long-term effects, including the long-term safety and durability of this therapy, are still lacking.

A third type of intervention in cases of SNAS is nickel chelation, but the data on this topic are scant. Chelation refers to the administration of a medication that binds to nickel (or other metal) and removes it from the body. Chelating agents reported to bind nickel as a treatment for SNAS are disulfuram, diethyldithiocarbamate, and EDTA disodium ethylenediamine tetra-acetate (EDTA). Of these, disulfuram has the best evidence, but not as monotherapy. Sharma reported results of a 2006 study in which 10 out of 11 patients had complete resolution of hand eczema with disulfuram combined with a low-nickel diet.⁴⁹ Both topical and oral preparations of EDTA are used for nickel allergy, with limited evidence behind the practice based on studies from the 1980s pertaining to the topical use of EDTA in nickel dermatitis.

The next type of intervention to consider in patients with SNAS are treatments to mitigate the allergic response to the systemic exposure to nickel. Systemic corticosteroids (prednisone, budesonide) are effective in acute flare-ups of SNAS such as in cases of a temporary, unintentional exposure to nickel. Their long-term side effects prohibit long-term use. Other traditional immunosuppressive treatments such as cyclosporine and methotrexate would be helpful in SNAS also, but there are limited data on the use of these treatments in SNAS. The newer agents for dermatitis are promising. Jacob et al reported on the success of treatment with off-label dupilumab in patients with systemic contact dermatitis to nickel or Balsam of Peru who were not strictly avoiding nickel in the diet.⁵⁰ Joshi et al reported on the use of dupilumab in a 44-year-old man with nickel allergy and numerous endovascular stents and vascular clips.⁵¹ There are no reports of intervention with dupilumab in nickel-related gastrointestinal symptomatology, but theoretically it may benefit some patients. There are yet any case reports of off-label use for upadacitinib, lebrikizumab, or tralokinumab in systemic nickel allergy syndrome. However, one could speculate that their use could be beneficial in certain treatment-resistant cases.

The final category of intervention in SNAS is the use of probiotics. Randazzo et al published a study in which patients with SNAS were administered *Lactobacillus reuteri* or placebo, and both the gastrointestinal symptoms and severity and frequency of cutaneous symptoms (urticaria, itch, and eczema) as well as recurrent abdominal pain improved in patients administered the *Lactobacillus* along with a low-nickel diet.⁵² Mori

et al reported a case of a patient with SNAS whose skin lesions and diarrhea improved with *Bifidobacterium* supplementation when low-nickel diet alone did not resolve the symptoms.⁵³ Lombardi et al reported the largest study on this topic: 51 patients were placed on a low-nickel diet and 22 of these were also given probiotics, either *Lactobacilli* or *Bifidobacteria* depending on the type of dysbiosis present.³⁵ The authors concluded that supplementation with dysbiosis-specific probiotics in the setting of nickel allergy greatly improves the outcome vs low-nickel diet alone.

Conclusion

Systemic nickel allergy syndrome (SNAS) is an underrecognized syndrome that has gastrointestinal, dermatologic, rheumatologic, and neurologic implications. Screening patients for possible food associations linked to their various symptoms or even just eliciting a history of metal contact allergy may be useful and easy ways to select patients who may benefit from patch testing. While a low-nickel diet is the most accepted primary intervention, there is an increasing body of case reports detailing the success of nickel hyposensitization, nickel chelation, systemic medications, or even probiotics. More studies are currently needed to validate these studies, but they may offer hope for patients with SNAS who have difficulty following such a restrictive diet.

Declaration of competing interest

None.

CRediT authorship contribution statement

Allison Freeman: Writing – original draft. **Ashleigh Letterman:** Writing – original draft. **Jeffrey Boos:** Writing – review & editing. **Scott Jackson:** Writing – original draft.

Authorship

The authors take full responsibility for the content of the article. All authors had access to the data and a role in writing this manuscript.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.ajmo.2026.100126>.

References

- Gangemi S, Ricciardi L, Minciullo PL, et al. Serum levels of protein oxidation products in patients with nickel allergy. *Allergy Asthma Proc.* 2009;30(5):552–557 PMID:19843408. doi:10.2500/aap.2009.30.3275.
- D'Alcamo A, Mansueto P, Soresi M, et al. Contact dermatitis due to nickel allergy in patients suffering from non-celiac wheat sensitivity. *Nutrients.* 2017;9(2):103 PMID:28157173. doi:10.3390/nu9020103.
- Minelli M, Schiavino D, Musca F, et al. Oral hyposensitization to nickel induces clinical improvement and a decrease in TH1 and TH2 cytokines in patients with systemic nickel allergy syndrome. *Int J Immunopathol Pharmacol.* 2010;23(1):193–201 PMID:20378005. doi:10.1177/039463201002300117.
- Erfani B, Lidén C, Midander K. Short and frequent skin contact with nickel. *Contact Dermatitis.* 2015;73(4):222–230 PMID:26086991. doi:10.1111/cod.12426.
- Ahlström MG, Midander K, Menné T, et al. Nickel deposition and penetration into the stratum corneum after short metallic nickel contact: an experimental study. *Contact Dermatitis.* 2019;80(2):86–93 PMID:30370609. doi:10.1111/cod.13136.
- Warshaw EM, Zhang AJ, DeKoven JG, et al. Epidemiology of nickel sensitivity: retrospective cross-sectional analysis of North American Contact Dermatitis Group data 1994–2014. *J Am Acad Dermatol.* 2019;80(3):701–713 PMID:30342160. doi:10.1016/j.jaad.2018.09.058.
- DeKoven JG, Warshaw EM, Reeder MJ, et al. North American contact dermatitis group patch test results: 2019–2020. *Dermatitis.* 2023;34(2):90–104 PMID:36917520. doi:10.1089/derm.2022.29017.jdk.
- Mukovozov IM, Kashetsky N, de Gannes G. Prevalence of contact allergy to nickel: a retrospective chart review. *Dermatitis.* 2022;33(5):355–361 PMID:36113000. doi:10.1097/DER.0000000000000812.
- Atwater AR, Ward JM, Liu B, et al. Contact allergy in Canada versus United States: analysis of the North American Contact Dermatitis Group data 2005–2016. *Dermatitis.* 2021;32(6):421–429 PMID:34238819. doi:10.1097/DER.0000000000000701.
- Ahlström MG, Thyssen JP, Wennervaldt M, Menné T, Johansen JD. Nickel allergy and allergic contact dermatitis: a clinical review of immunology, epidemiology, exposure, and treatment. *Contact Dermatitis.* 2019;81(4):227–241 PMID:31140194. doi:10.1111/cod.13327.
- Svedman C, Ale I, Goh CL, et al. Patch testing with nickel sulfate 5.0% traces significantly more contact allergy than 2.5%: a prospective study within the International Contact Dermatitis Research Group. *Dermatitis.* 2022;33(6):417–420 PMID:36125779. doi:10.1097/DER.0000000000000935.
- Conti MV, Bissacco G, De Giuseppe R, et al. Systemic nickel allergy syndrome (SNAS): taking stock of medical nutrition therapy SNAS and nutrition. *J Community Med Public Health.* 2021;5:225. doi:10.29011/2577-2228.100225.
- Josefson A, Färm G, Meding B. Validity of self-reported nickel allergy. *Contact Dermatitis.* 2010;62(5):289–293 PMID:20536476. doi:10.1111/j.1600-0536.2010.01702.x.
- Chung M, Bandemer B, Cafaro T, LaRiccia P. Refractory systemic nickel allergy syndrome management: 5 cases and chart review survey from an integrative medical practice. *Integ Comple Therap.* 2023;29(4):176–181. doi:10.20944/preprints202304.0185.v1.
- Guérout AM, Al-Balah A, Davies AH, Shalhoub J. Nickel hypersensitivity and endovascular devices: a systematic review and meta-analysis. *Heart.* 2022;108(21):1707–1715 PMID:34702756. doi:10.1136/heartjnl-2021-319940.
- Kocayigit İ, Can Y, Şahinöz M, Saçlı H, Kara İ. Surgical removal of atrial septal defect closure device due to systemic allergic reaction to nickel: a case report. *Türk Gogus Kalp Damar Cerrahisi Derg.* 2022;30(4):631–634 PMID:36605319. doi:10.5606/tgkdc.dergisi.2022.23893.
- Nosbaum A, Rival-Tringali AL, Barth X, et al. Nickel-induced systemic allergic dermatitis from a sacral neurostimulator. *Contact Dermatitis.* 2008;59(5):319–320 PMID:18976388. doi:10.1111/j.1600-0536.2008.01434.x.
- Giraldo-Tugores M, Sanmartín-Fernández M, Fernández-Lozano C, Martínez-Botas J, De-la-Hoz-Caballer B, González-de-Olano D. Allergic contact dermatitis from nickel and coronary aneurysm. *Contact Dermatitis.* 2023;88(6):480–482 PMID:36799484. doi:10.1111/cod.14295.
- Kounis NG, Kounis GN, Koutsojannis C, Tsigkas G, Almpanis G, Mazarakis A. Metal allergy, atrial septal occluder devices and the risk of Kounis syndrome. *Ann Thorac Surg.* 2010;90(6):2087–2088 PMID:21095381. doi:10.1016/j.athoracsurg.2010.04.076.
- Apostolos A, Gregoriou S, Drakopoulou M, et al. Patent foramen ovale closure in patients with and without nickel hypersensitivity: a randomized trial. *Circ Cardiovasc Interv.* 2025;18(4):e015228. doi:10.1161/circinterventions.125.015228.
- Cole SC, Eftekhari K, Anderson RL, Oberg TJ. Systemic allergic response in the setting of a metallic intraorbital foreign body with intraoperative magnet-assisted retrieval. *Ophthalmic Plast Reconstr Surg.* 2017;33(4):e102–e104 PMID:27811635. doi:10.1097/IOP.0000000000000819.
- McLean L, Yewchuk L, Israel DM, Prendiville JS. Acute onset of generalized pruritic rash in a toddler. Diagnosis: systemic allergic (contact) dermatitis to nickel from ingestion of metal coins. *Pediatr Dermatol.* 2011;28(1):53–54 PMID:21276052. doi:10.1111/j.1525-1470.2010.01367.x.
- Ricciardi L, Arena A, Arena E, et al. Systemic nickel allergy syndrome: epidemiological data from four Italian allergy units. *Int J Immunopathol Pharmacol.* 2014;27(1):131–136 PMID:24674689. doi:10.1177/039463201402700118.
- de Waal D. Nickel allergy masquerading as irritable bowel syndrome: case report. *J Acad Nutr Diet.* 2026;126(1):156208 PMID:41043685. doi:10.1016/j.jand.2025.09.010.
- Cazzato IA, Vadrucchi E, Cammarota G, Minelli M, Gasbarrini A. Lactose intolerance in systemic nickel allergy syndrome. *Int J Immunopathol Pharmacol.* 2011;24(2):535–537 PMID:21658331. doi:10.1177/039463201102400230.
- Stanghellini V, Tosetti C, Benedetto E, et al. Nickel sensitization in patients with gastro-esophageal reflux disease. *United Eur Gastroenterol J.* 2015;4(2):184–190. doi:10.1177/2050640615595917.
- Zirvas MJ, Molenda MA. Dietary nickel as a cause of systemic contact dermatitis. *J Clin Aesthet Dermatol.* 2009;2(6):39–43 PMID:20729949.
- Silvestri DL, Barmettler S. Pruritus ani as a manifestation of systemic contact dermatitis: resolution with dietary nickel restriction. *Dermatitis.* 2011;22(1):50–55 PMID:21291644.
- Kolodziej T, Szepletowski JC, Sikora J, Bialynick-Birula R. The baboon syndrome due to nickel. *Acta Dermatovenerol Croat.* 2003;11(1):29–31 PMID:12718793.
- Mancuso G, Berdondini RM. Eyelid dermatitis and conjunctivitis as sole manifestations of allergy to nickel in an orthodontic appliance. *Contact Dermatitis.* 2002;46(4):245 PMID:12081709. doi:10.1034/j.1600-0536.2002.460415.x.
- Braga M, Quecchia C, Perotta C, et al. Systemic nickel allergy syndrome: nosologic framework and usefulness of diet regimen for diagnosis. *Int J Immunopathol Pharmacol.* 2013;26(3):707–716 PMID:24067467. doi:10.1177/039463201302600314.
- Fernandes P, Sharma SR, Magee A, Michielon G, Fraise A. Severe migraine associated with nickel allergy requiring surgical removal of atrial septal device. *Ann Thorac Surg.* 2019;108(3):e183–e184 PMID:30794784. doi:10.1016/j.athoracsurg.2019.01.034.
- Loyo E, Jara LJ, López PD, Puig AC. Autoimmunity in connection with a metal implant: a case of autoimmune/autoinflammatory syndrome induced by adjuvants. *Auto Immun Highlights.* 2012;4(1):33–38 PMID:26000140. doi:10.1007/s13317-012-0044-1.
- Stejskal V, Reynolds T, Björklund G. Increased frequency of delayed type hypersensitivity to metals in patients with connective tissue disease. *J Trace Elem Med Biol.* 2015;31:230–236 PMID:25636536. doi:10.1016/j.jtemb.2015.01.001.
- Lombardi F, Fiasca F, Minelli M, et al. The effects of low-nickel diet combined with oral administration of selected probiotics on patients with systemic nickel allergy

- syndrome (SNAS) and gut dysbiosis. *Nutrients*. 2020;12(4):1040 PMID:32283870. doi:10.3390/nu12041040.
36. Kolberg L, Forster F, Gerlich J, et al. Nickel allergy is associated with wheezing and asthma in a cohort of young German adults: results from the SOLAR study. *ERJ Open Res*. 2020;6(1):00178–02019 PMID:32039258. doi:10.1183/23120541.00178-2019.
 37. Watanabe M, Masieri S, Costantini D, et al. Overweight and obese patients with nickel allergy have a worse metabolic profile compared to weight matched non-allergic individuals. *PLoS One*. 2018;13(8):e0202683 PMID:30153310. doi:10.1371/journal.pone.0202683.
 38. Lusi EA, Di Ciommo VM, Patritti T, Guarascio P. High prevalence of nickel allergy in an overweight female population: a pilot observational analysis. *PLoS One*. 2015;10(3):e0123265 PMID:25822975. doi:10.1371/journal.pone.0123265.
 39. Risi R, Masieri S, Poggiogalle E, et al. Nickel sensitivity is associated with GH-IGF1 axis impairment and pituitary abnormalities on MRI in overweight and obese subjects. *Int J Mol Sci*. 2020;21(24):9733 PMID:33419306. doi:10.3390/ijms21249733.
 40. Andrioli M, Trimboli P, Maio D, Persani L, Minelli M. Systemic nickel allergic syndrome as an immune-mediated disease with an increased risk for thyroid autoimmunity. *Endocrine*. 2015;50(3):807–810. doi:10.1007/s12020-015-0581-2.
 41. Yuk JS, Shin JS, Shin JY, Oh E, Kim H, Park WI. Nickel allergy is a risk factor for endometriosis: an 11-year population-based nested case-control study. *PLoS One*. 2015;10(10):e0139388. doi:10.1371/journal.pone.0139388.
 42. Minelli M, Schiavino D, Musca F, et al. Oral hyposensitization to nickel induces clinical improvement and a decrease in TH1 and TH2 cytokines in patients with systemic nickel allergy syndrome. *Int J Immunopathol Pharmacol*. 2010;23(1):193–201 PMID:20378005. doi:10.1177/039463201002300117.
 43. Manfredi A, Varrà MO, Zanardi E, et al. Dietary exposure assessment to nickel through the consumption of poultry, beef, and pork meat for different age groups in the Italian population. *Ital J Food Saf*. 2025;14(4):13840 PMID:40824104. doi:10.4081/ijfs.2025.13840.
 44. Da Mata Perez L, França AT, Zimmerman JR. Systemic nickel allergy syndrome. *World Allergy Organ J*. 2015;8(Suppl 1):A89 PMID:24406458. doi:10.1186/1939-4551-8-S1-A89.
 45. Rizzi A, Nucera E, Laterza L, et al. Irritable bowel syndrome and nickel allergy: what is the role of the low nickel diet? *J Neurogastroenterol Motil*. 2017;23(1):101–108 PMID:28049864. doi:10.5056/jnm16027.
 46. Nucera E, Rizzi A, Chini R, et al. Diet intervention study through telemedicine assistance for systemic nickel allergy syndrome patients during the COVID-19 pandemic. *Nutrients*. 2021;13(8):2897. doi:10.3390/nu13082897.
 47. Rizzi A, Chini R, Porcari S, et al. Datterino trial: a double blind, randomized, controlled, crossover, clinical trial on the use of hydroponic cultivated tomato sauce in systemic nickel allergy syndrome. *J Clin Med*. 2022;11(18):5459. doi:10.3390/jcm11185459.
 48. Di Gioacchino M, Ricciardi L, De Pità O, et al. Nickel oral hyposensitization in patients with systemic nickel allergy syndrome. *Ann Med*. 2014;46(1):31–37 PMID:24256166. doi:10.3109/07853890.2013.861158.
 49. Sharma AD. Disulfiram and low nickel diet in the management of hand eczema: a clinical study. *Indian J Dermatol Venereol Leprol*. 2006;72(2):113–118 PMID:16707816. doi:10.4103/0378-6323.25635.
 50. Jacob SE, Sung CT, Machler BC. Dupilumab for systemic allergy syndrome with dermatitis. *Dermatitis*. 2019;30(2):164–167 PMID:30829809. doi:10.1097/DER.0000000000000446.
 51. Joshi SR, Khan DA. Effective use of Dupilumab in managing systemic allergic contact dermatitis. *Dermatitis*. 2018;29(5):282–284 PMID:30179979. doi:10.1097/DER.0000000000000409.
 52. Randazzo CL, Pino A, Ricciardi L, et al. Probiotic supplementation in systemic nickel allergy syndrome patients: study of its effects on lactic acid bacteria population and on clinical symptoms. *J Appl Microbiol*. 2015;118(1):202–211 PMID:25363062. doi:10.1111/jam.12685.
 53. Mori H, Hayashi S, Mori S, et al. A case of systemic nickel allergy with diarrhea-predominant irritable bowel syndrome in which nickel intake restriction and administration of a probiotic formulation were effective. *Allergol Int*. 2021;70(4):515–516 PMID:33853743. doi:10.1016/j.alit.2021.03.004.